Hongyi Jing

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EDUCATION

Huazhong University of Science and Technology (HUST), Wuhan, China September 2020 - June 2024
Bachelor of Engineering, School of Artificial Intelligence and Automation
Major: Artificial Intelligence; **Cumulative GPA:** 3.90/4.0

PUBLICATIONS

- **Hongyi Jing**, Xianjing Huang, and Xinglin Guo. "Semi-supervised learning on Lightweight OpenPose", *ETAI 2022*, accepted.
- Yuwen Tan, Xiang Xiang, Yifeng Chen, **Hongyi Jing**, Shiyang Ye, Chaoran Xue and Hui Xu. "Coupling Bracket Segmentation and Tooth Surface Reconstruction on 3D Dental Models", *MICCAI 2023*, published.

RESEARCH EXPERIENCES

Large Language Model in Chemistry July 2023 - September 2023
Research Assistant; Advisor: Prof. Xiangliang Zhang from Notre Dame
Lab: Machine Intelligence and kNowledge Engineering (MINE) Lab[1], University of Notre Dame

- Explore the application of large language models (LLMs) in solving diverse chemical problems, including reaction prediction, relation extraction, and named entity recognition;
- Used prompts and chain of thought to improve accuracy in chemical reaction prediction using LLMs; explored prompt tuning and LangChain methods to access the capabilities of LLMs in predicting products;
- Evaluated the capability boundaries of LLMs in chemical text extraction tasks, comparing their performance against specialized models to analyze the advantages and shortcomings of LLMs in chemical tasks;
- Identified optimal chemical datasets for LLMs to complete specific chemical tasks, and establish a comprehensive system for evaluating the chemical capabilities of LLMs.

3D Dental Model Segmentation and Reconstruction August 2022 - February 2023
Research Assistant; Advisor: Prof. Xiang Xiang from HUST
Lab: Human-AI Interaction and Visualization (HAIV) Lab[2], HUST

- Employed deep learning networks and projection-based surface reconstruction to develop an efficient approach for tooth segmentation and 3D reconstruction in orthodontic treatment.
- Designed a graph-based network for bracket segmentation on 3D dental models, employing dilated k-nearest neighbors (knn) and residual blocks for comprehensive feature extraction and integration.
- Performed post-processing after bracket segmentation to optimize tooth edges, leveraging angular relationships to remove inaccuracies and enhance surface smoothness.
- Participated in 3D dental model reconstruction post-bracket removal by projecting the dental surface onto a 2D plane, repairing it based on curvature information, and re-projecting it onto the original 3D tooth surface;
- Compared with other commonly used open-source surface repair algorithms that do not consider curvature information, ours preserves the fine details of tooth surface curvature while staying time-efficient.

Semi-Supervised Learning on Lightweight Openpose June 2022 - August 2022
Research Leader; Advisor: Prof. Rabih Younes from Duke University

- Researched semi-supervised human pose estimation with lightweight OpenPose using heatmaps and part affinity; applied two novel methods and achieved comparative performance with only half of the annotated data;
- Applied strong and weak augmentations for input unlabelled images to the task of HPE referring to FixMatch; incorporated a threshold mechanism to filter inaccurate pseudo labels and guide accurate learning;
- Adopted the ST++ semi-supervised learning architecture in HPE, categorizing unlabelled images into two categories by prediction difficulty, and prioritizing learning from simple images before challenging ones.
- Compared our methods with the baseline, ours achieve comparable accuracy with just half of the labeled information, and our semi-supervised method with the threshold mechanism sometimes even surpasses the results of fully supervised method using all annotated data.

ADDITIONAL INFORMATION

- Programming: Python, C, C++, Matlab.
- Research Interests: Deep learning in computer vision, 3D reconstruction, 3D Vision, Multimodal learning.
- Languages: Toefl 111(Reading 30, Listening 30, Speaking 23, Writing 28)

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[1] Official website of MINE Lab is available at: https://sites.nd.edu/xiangliang-zhang/
[2] Official website of HAIV Lab is available at: https://haivlab.wixsite.com/home